



Database

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Relational vs Non – Relational

	Relational Database	Non-Relational Database
Data Integrity	Enforces data integrity through relationships and constraints	Flexible schema allows for easy data updates
Structured Queries	Supports complex queries using SQL	May offer simpler, faster query performance
Scalability	May not scale as well for large datasets	Highly scalable and can handle big data and high traffic
ACID Transactions	Provides ACID (Atomicity, Consistency, Isolation, Durability) transactions	May sacrifice ACID properties for improved performance or scalability
Consistency	Ensures strong data consistency	May offer eventual consistency or tunable consistency
Schema Evolution	Schema changes can be complex and time-consuming	Allows for easy schema changes and flexibility in data model
Examples	MySQL, PostgreSQL, Oracle	MongoDB, Cassandra, Redis, DynamoDB



Relational Database

This lecture assume you already pass IT004 and have knowledge of SQL Queries

See [SQL Tutorial \(w3schools.com\)](http://w3schools.com) if you need revision

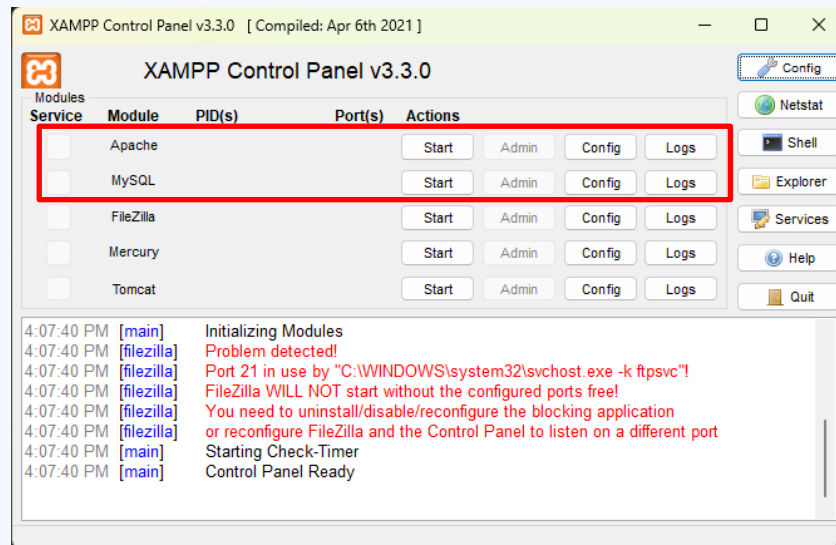


MySQL

This course will use MySQL as Database Server and PHPMyadmin as DBMS

Install Xampp: [XAMPP Installers and Downloads for Apache Friends](#)

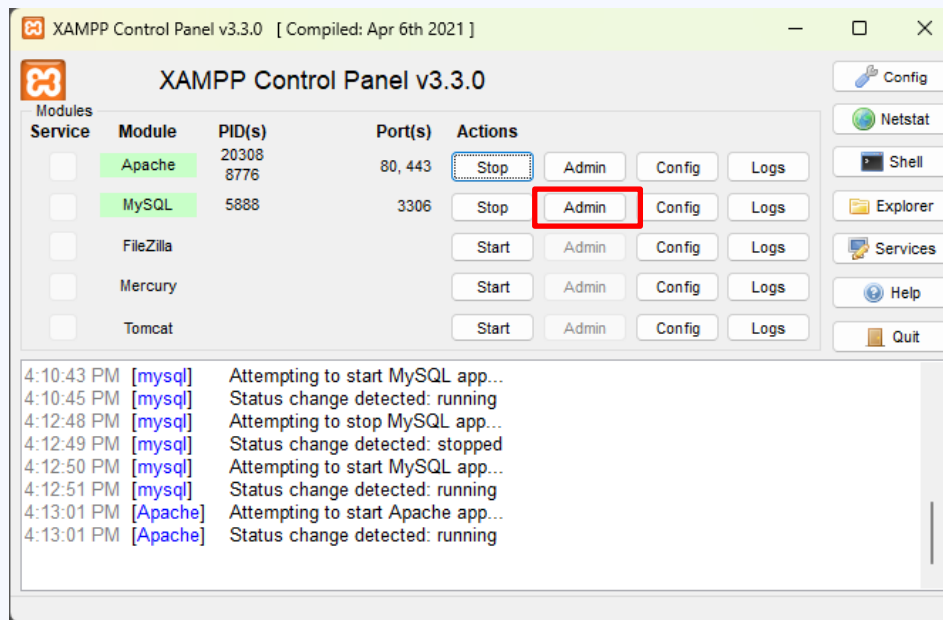
Start both PHP and MySQL Server and you're ready to go





MySQL

Click on Admin to open phpMyAdmin DBMS site on your local machine





MySQL

The screenshot displays the phpMyAdmin web interface for a MySQL server. The top navigation bar includes tabs for Databases, SQL, Status, User accounts, Export, Import, Settings, Replication, Variables, Charsets, Engines, and Plugins. The left sidebar shows a tree view of databases, including 'New', 'information_schema', 'mysql', 'performance_schema', 'phpmyadmin', 'pv', 'test', and 'wordpress_enhance_ui'. The main content area is divided into several sections:

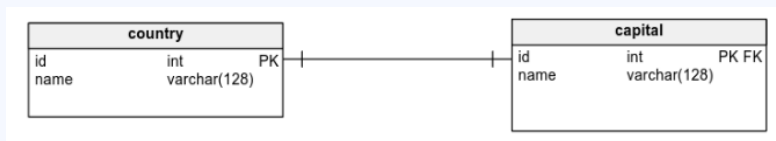
- General settings:** Displays 'Server connection collation' set to 'utf8mb4_unicode_ci' with a 'More settings' link.
- Appearance settings:** Shows 'Language' set to 'English' and 'Theme' set to 'pmahomme' with a 'View all' button.
- Database server:** Lists server details: Server: 127.0.0.1 via TCP/IP, Server type: MariaDB, Server connection: SSL is not being used, Server version: 10.4.27-MariaDB - mariadb.org binary distribution, Protocol version: 10, User: root@localhost, and Server charset: UTF-8 Unicode (utf8mb4).
- Web server:** Lists web server details: Apache/2.4.54 (Win64) OpenSSL/1.1.1p PHP/8.2.0, Database client version: libmysql - mysqlnd 8.2.0, PHP extension: mysqli, curl, mbstring, and PHP version: 8.2.0.
- phpMyAdmin:** Provides version information (5.2.0, latest stable version: 5.2.1) and links to Documentation, Official Homepage, Contribute, Get support, List of changes, and License.

A notification banner at the bottom states: 'A newer version of phpMyAdmin is available and you should consider upgrading. The newest version is 5.2.1, released on 2023-02-08.'



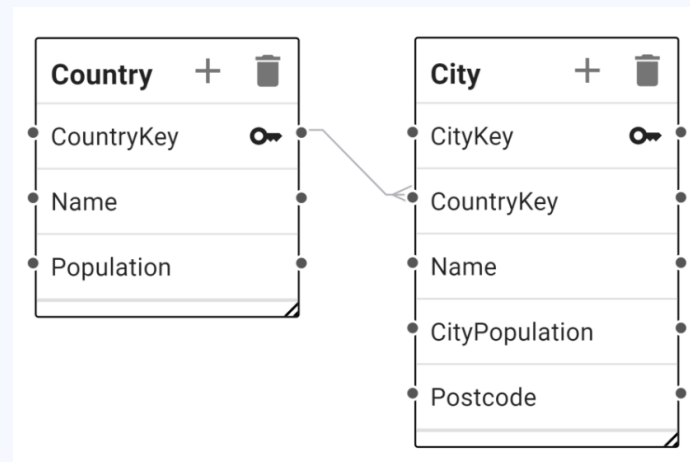
1 – 1, 1 – n, n – n Relationship

Primary key ref Primary key



1 – 1 relationship

Foreign key ref Primary key

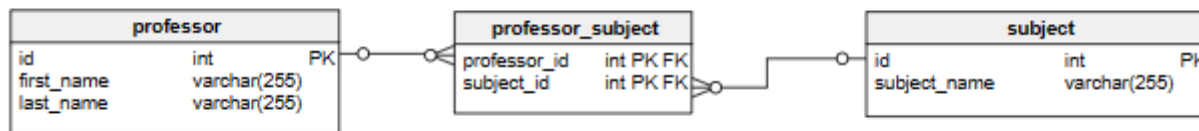


1 – n relationship



1 – 1, 1 – n, n – n Relationship

Use intermediate table



n – n relationship



Object Relational Mapping (ORM)

Instead of writing raw queries, we can map a **table** with a **model**

Most frameworks now support ORM, allow us to interact with database via Object

ORM provide simpler syntax, more readable than raw queries



Example: PHP Laravel

```
class User extends Model {  
  
    protected $table = 'my_users';  
  
}
```

Laravel already cover all the setup we need in the **Model** class

Now we can interact with `my\_users` table via **User** class

Eloquent ORM - Laravel 5.0 - The PHP Framework For Web Artisans



Example: PHP Laravel

```
$users = User::all(); // get all records  
$user = User::find(1); //find record with primary key value is 1  
$model = User::where('votes', '>', 100)->firstOrFail(); // complex queries  
$user->save(); // update
```

[Overview of Entity Framework Core - EF Core | Microsoft Learn](#)

[Sequelize | Feature-rich ORM for modern TypeScript & JavaScript](#)

[Eloquent ORM - Laravel 5.0 - The PHP Framework For Web Artisans](#)



Non – Relational Database (NoSQL Database)

Storing data as **Collections** and **Documents** (just like **tables** and **records**)

More Flexible Schema than Relational Database

Syntax similar to ORM



MongoDB

Use Free Service to get start [MongoDB Atlas](#) | [MongoDB](#)

DEPLOYMENT

Database

Data Lake

SERVICES

Device Sync

Triggers

Data API

Data Federation

Atlas Search

Stream Processing

Migration

SECURITY

Quickstart

Backup

Database Access

Network Access

Advanced

New On Atlas 2

Goto

Database Deployments

Find a database deployment...

Edit Config + Create

Cluster0 Connect View Monitoring Browse Collections ... FREE SHARED

Your cluster has been automatically paused due to prolonged inactivity.
Resume your cluster to connect to it and to gain access to your data.

Resume

VERSION	REGION	CLUSTER TIER	TYPE	BACKUPS	LINKED APP SERVICES	ATLAS SEARCH
7.0.6	AWS / Singapore (ap-southeast-1)	M0 Sandbox (General)	Replica Set - 3 nodes	Inactive	None Linked	Create Index

+ Add Tag



MongoDB

Get Connection string and connect with nodejs

[Node.js MongoDB Create Database \(w3schools.com\)](#)

Connecting with MongoDB Driver

1. Select your driver and version

We recommend installing and using the latest driver version.

Driver

Version

Node.js

5.5 or later

2. Install your driver

Run the following on the command line

```
npm install mongodb
```

[View MongoDB Node.js Driver installation instructions.](#)

3. Add your connection string into your application code

☐ View full code sample

```
mongodb+srv://fkmddev:<password>@cluster0.vnqljo6.mongodb.net/?  
retryWrites=true&w=majority&appName=Cluster0
```



Firestore

[Get started with Cloud Firestore | Firebase \(google.com\)](#)



So sánh và Thao tác – MySQL và MongoDB

Yêu cầu:

- 1. Tạo bảng users trong MySQL (sử dụng PHPMyAdmin):
 - Cột id (INT, AUTO_INCREMENT, PRIMARY KEY)
 - Cột name (VARCHAR)
 - Cột email (VARCHAR)
- 2. Tạo collection users trong MongoDB Atlas:
 - Mỗi document có các trường name, email
- 3. Thêm ít nhất 3 bản ghi cho mỗi loại cơ sở dữ liệu với dữ liệu giống nhau.
- 4. Chụp ảnh màn hình kết quả (bảng hoặc document hiển thị).
- 5. So sánh và trả lời các câu hỏi sau (viết tay hoặc file txt):
 - Cấu trúc dữ liệu trong MySQL và MongoDB có gì giống và khác nhau?
 - Nếu bạn cần linh hoạt trong việc thay đổi cấu trúc dữ liệu, bạn chọn MySQL hay MongoDB? Vì sao?
 - Hãy nêu một ví dụ ứng dụng thực tế phù hợp với MongoDB hơn MySQL.